

Technology Stack

Date	15 March 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, Vanilla JS	Lightweight, responsive glassmorphism UI; no build step needed.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.10 + Flask	Fast to build REST API; integrates directly with the ML model.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV dataset + pickled model	Crop_recommendation.csv for training; model.pkl for inference.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Vercel (serverless)	Free, auto-scaling deployment straight from GitHub.
5	Version Control & CI/CD	Git + GitHub	Version control and team collaboration.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	scikit-learn, Pandas, NumPy	Logistic Regression & K-Means; data processing and arrays.